AI for Manufacturing GRCC Training

AI for Workforce

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Welcome!

Get all the content at:
https://github.com/grcc-resendiz/ai-manufacturing-muskegon

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Day 3 Agenda

- **0:00** Welcome and Day 2 recap
- **0:10** Why downtime matters and how predictive maintenance works
- **0:20** Machine learning basics and how to judge a model
- **0:50** Hands on: predict machine failures in Orange
- **1:35** Break
- **1:45** Demo: the same model in Python
- **2:00** Implementation: AI readiness and your first pilot
- **2:45** Key takeaways and questions

Predictive Maintenance with Machine Learning

Using Data to Predict Equipment Failure



The Problem (manufacturing)

Cost of Downtime[1]

Unplanned downtime costs the world's 500 largest companies about $1.4 trillion a year.

Cost per Hour[1]

A large automotive plant loses about $2.3 million for every hour it stands still.

The Solution – Predictive Maintenance

Definition

Predict failures *before* they happen using data.

How it works

- Collect data from machines (temperature, vibration, etc.)
- Analyze the data to find patterns
- Predict when a failure is likely and take action.

Impact[3]

27

Downtime Hours

Per plant each month in 2024, down from 39 in 2019

2x

More PdM Teams

Almost half of firms now have predictive maintenance teams





Machine Learning (ML)

Machine learning is a branch of artificial intelligence that focuses on building systems that learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for each task.

The three major areas of ML are:

1

Supervised Learning

The model learns to make predictions from labeled data, where the correct output (label) is known. **Ex:** predicting machine failure

2

Unsupervised Learning

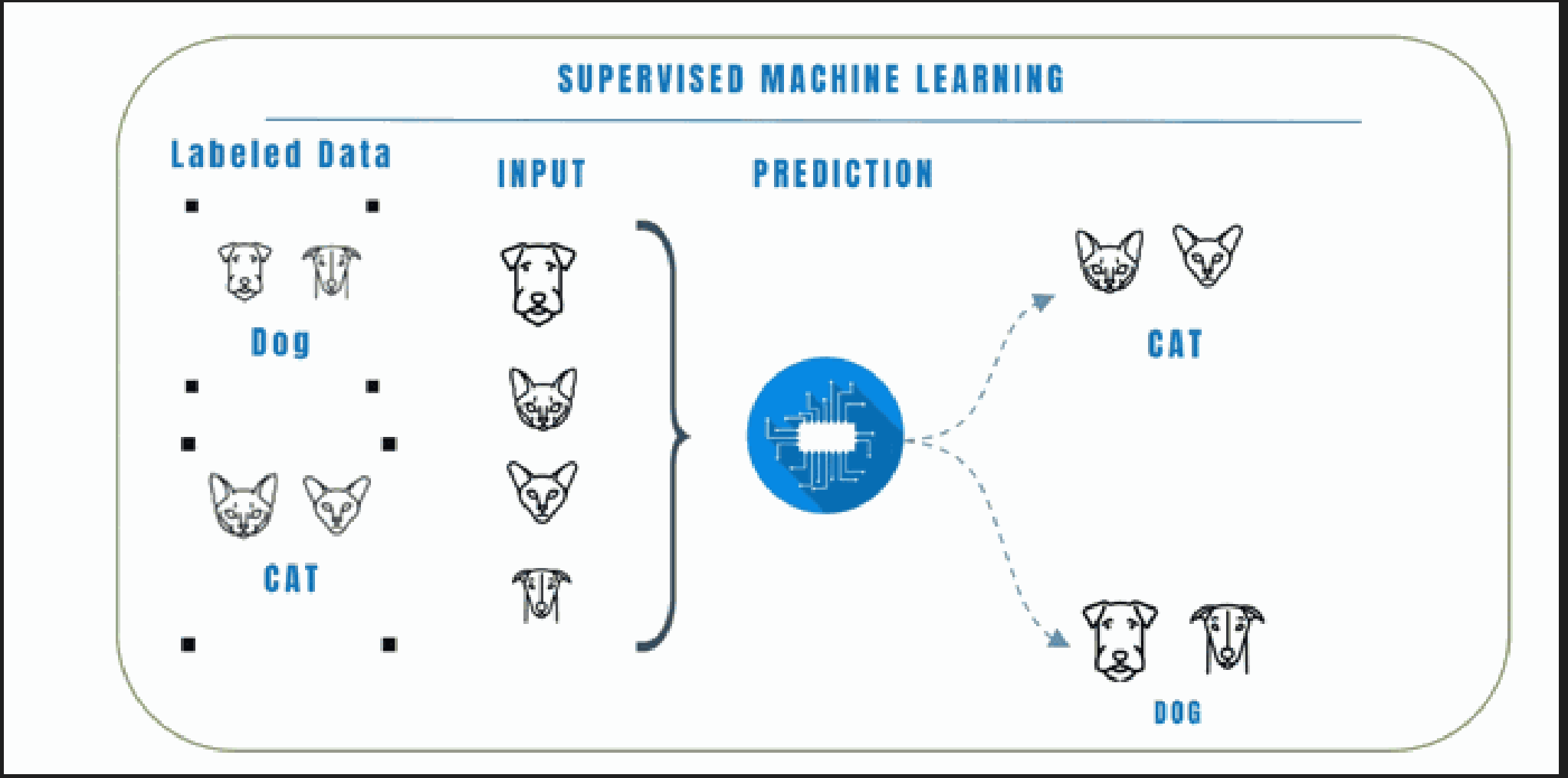
The model learns from unlabeled data, without predefined outputs. **Ex:** grouping similar machines

3

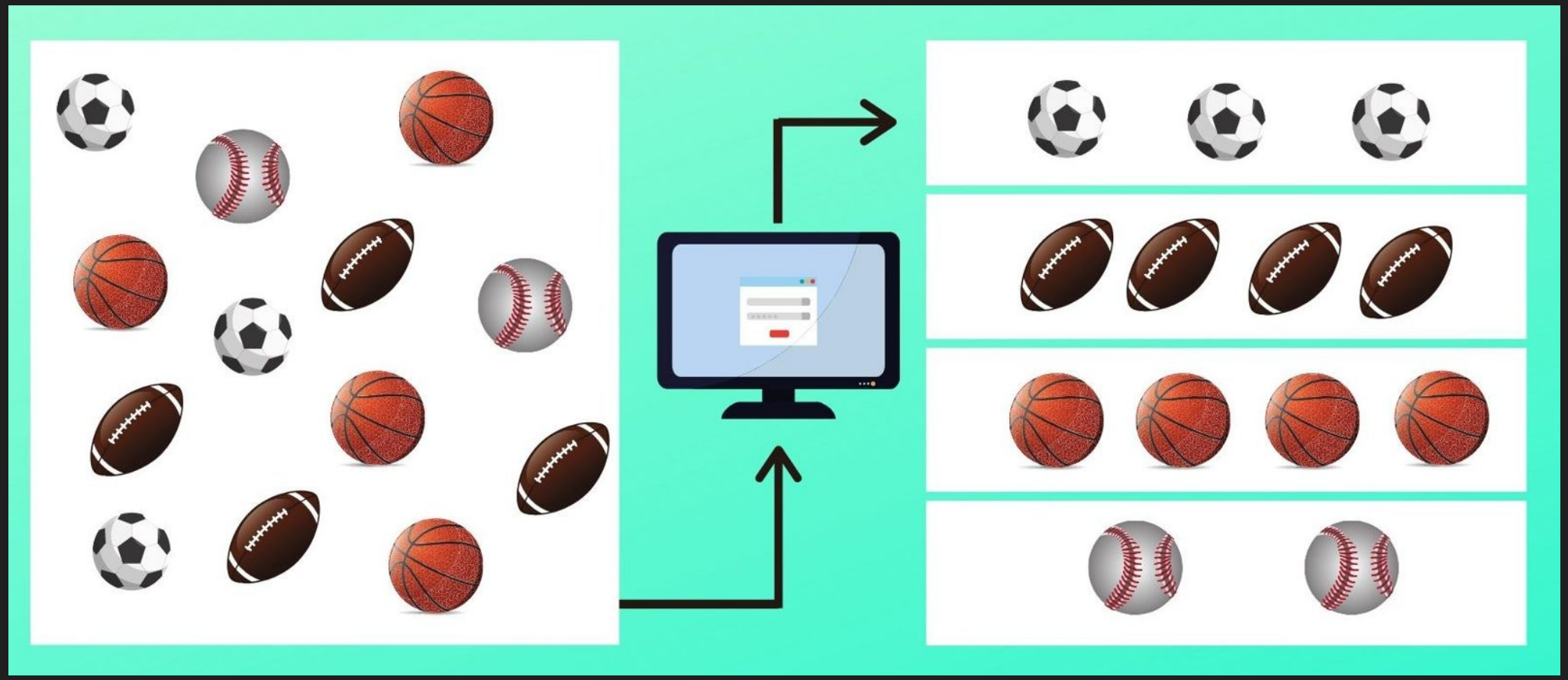
Reinforcement Learning

The model learns by interacting with an environment and receiving rewards or penalties. **Ex:** game-playing AI such as AlphaGo

Supervised Learning Classification Example



Unsupervised Clustering Example

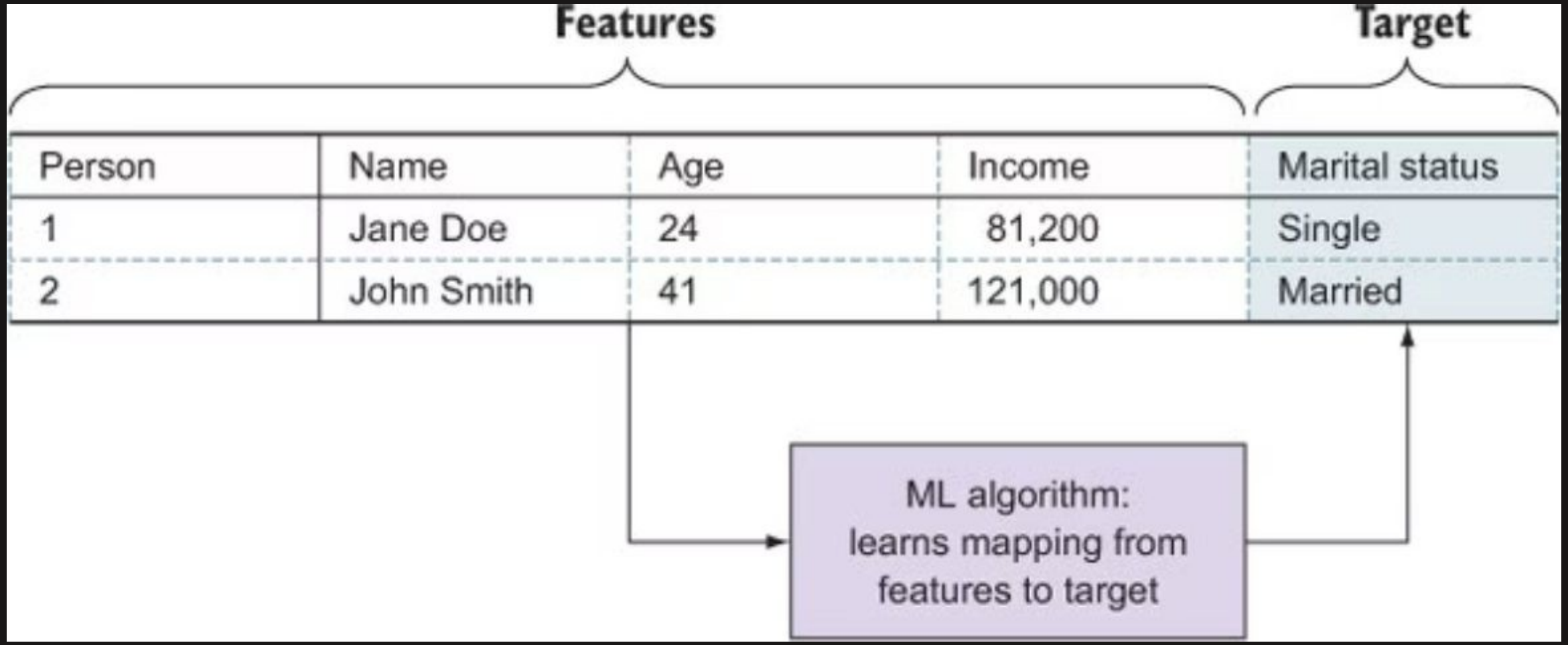




Reinforcement example – Pac-



Supervised Learning with Statistical Data



Key Machine Learning Terms

Training

The process of using data to teach the model patterns

Prediction

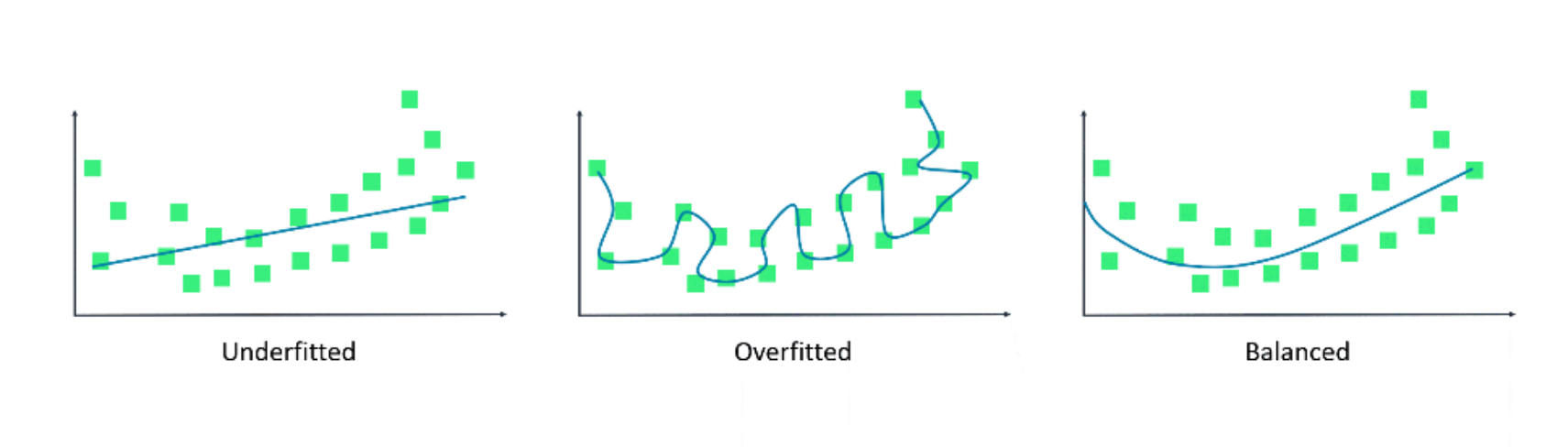
The output produced by a trained model on new data

Overfitting

When a model learns the training data too well and performs poorly on new data

Underfitting

When a model is too simple to capture patterns in the data



Machine Learning & Supervised Learning

What is Machine Learning?

- Allowing a machine to learn patterns from data (pre-set algorithms)
- Use those patterns to make predictions

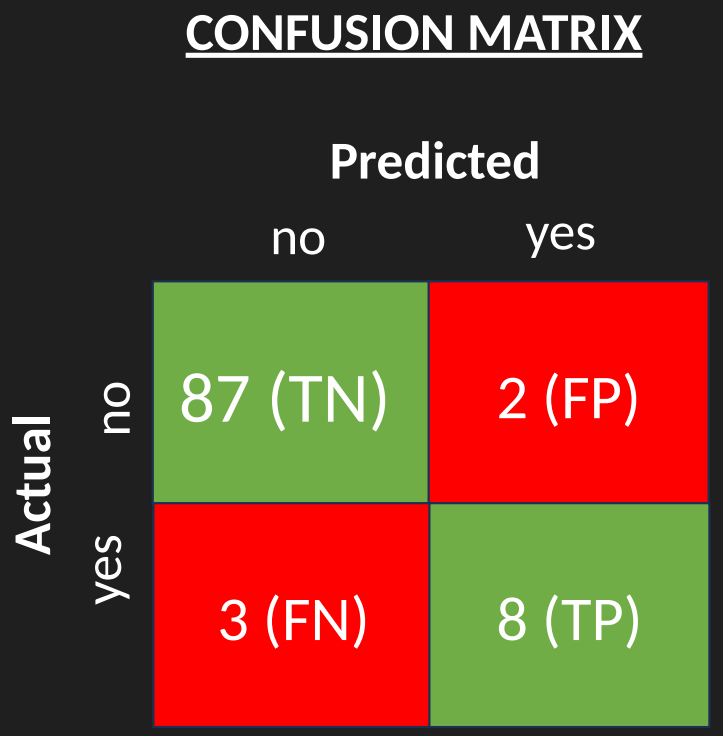
What is Supervised Learning?

- A type of machine learning that learns from labeled data
- Data includes: **Inputs** (machine measurements) and **Outputs** or **Labels** (failure or no failure)

Example (simplified data)

Temperature (K)	Vibration (mm/s)	Failure (0 or 1)
308.5	4.3	0
310.0	4.2	1
309.2	2.5	0
310.7	3.9	1
311.3	2.4	0
308.9	3.8	0
311.5	4.3	1
310.1	2.2	0

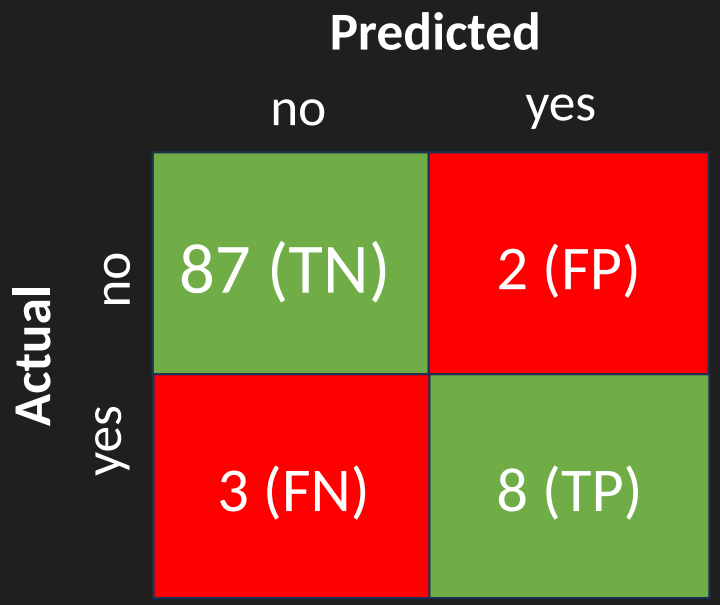
Confusion Matrix



- A **Confusion Matrix** compares predicted labels to actual labels.
- Suppose a model predicts whether a machine will fail during the next shift, across 100 machine shifts.
  - **87, True Negatives (TN):**
    Healthy machines correctly predicted to keep running.
  - **2, False Positives (FP):**
    False alarms: predicted a failure, but the machine was fine.
  - **3, False Negatives (FN):**
    Missed failures: predicted fine, but the machine failed.
  - **8, True Positives (TP):**
    Failures caught before they happened.

Confusion Matrix Evaluation Metrics (No Calculations)

CONFUSION MATRIX



- **Accuracy** tells us the proportion of all predictions that are correct.
  - Ex: Out of 100 machine shifts, what percent did the model get right?
- **Precision** focuses on minimizing false positives. It finds the proportion of positive predictions that are correct.
  - Ex: Of the 10 failure warnings, how many were real failures?
- **Recall** focuses on minimizing false negatives. It finds the proportion of true positives that are correctly identified.
  - Ex: Of the 11 real failures, how many did the model catch?
- **F1 Score** balances precision and recall into a single metric.



Confusion Matrix Evaluation Metrics (Calculations)

CONFUSION MATRIX

Actual \ Predicted	no	yes
no	87 (TN)	2 (FP)
yes	3 (FN)	8 (TP)

- **Accuracy** = (TP + TN) / Total = (8 + 87) / 100 = 95%
  - 95 of 100 predictions were right.
- **Precision** = TP / (TP + FP) = 8 / (8 + 2) = 80%
  - When the model warned of a failure, it was right 80% of the time.
- **Recall** = TP / (TP + FN) = 8 / (8 + 3) = about 73%
  - The model caught 73% of real failures. It missed 3.
- **F1 Score** = 2 x (0.80 x 0.73) / (0.80 + 0.73) = about 76%
  - One score that balances catching failures and avoiding false alarms.

Introduction to Orange Data Mining

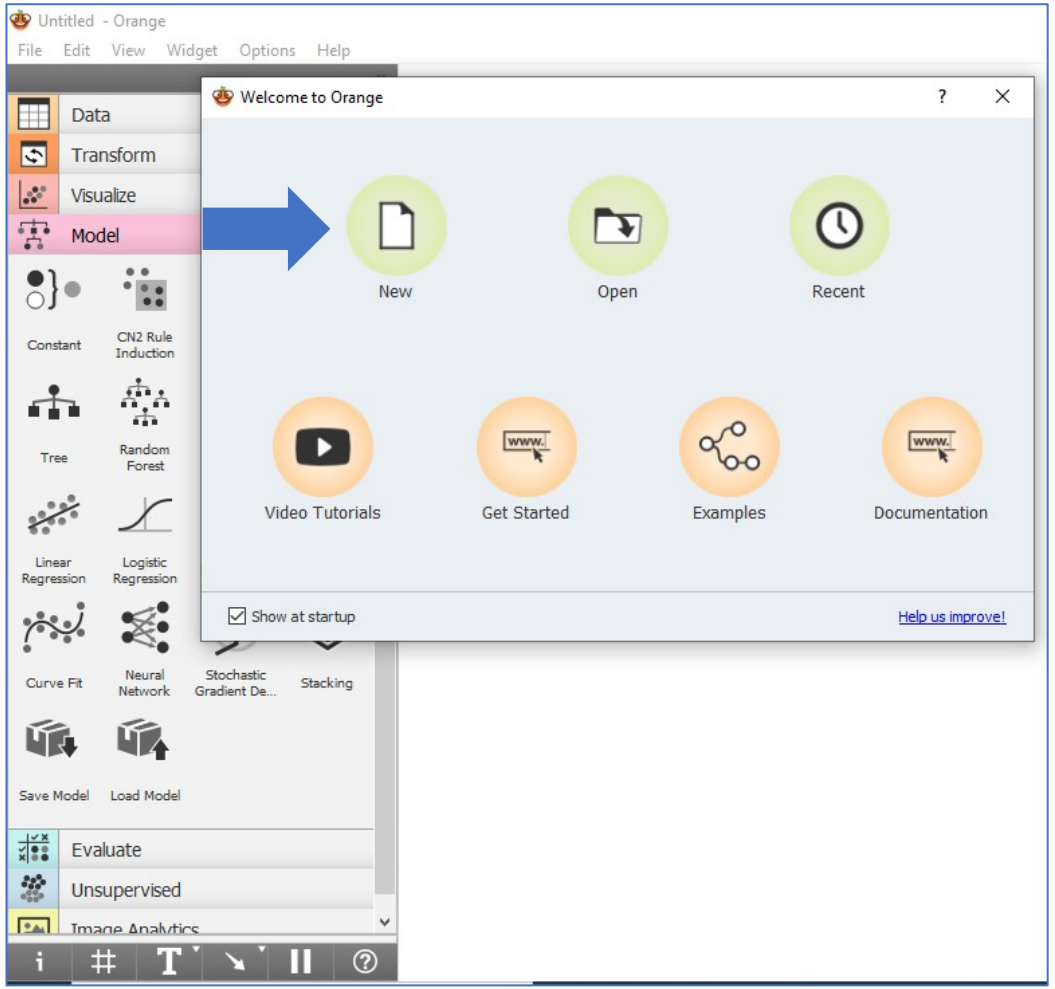
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Introduction to Orange Data Mining

1. Install Orange before class: orangedatamining.com/download
2. Getting Started with Orange Data Mining
3. Exploring the Canvas

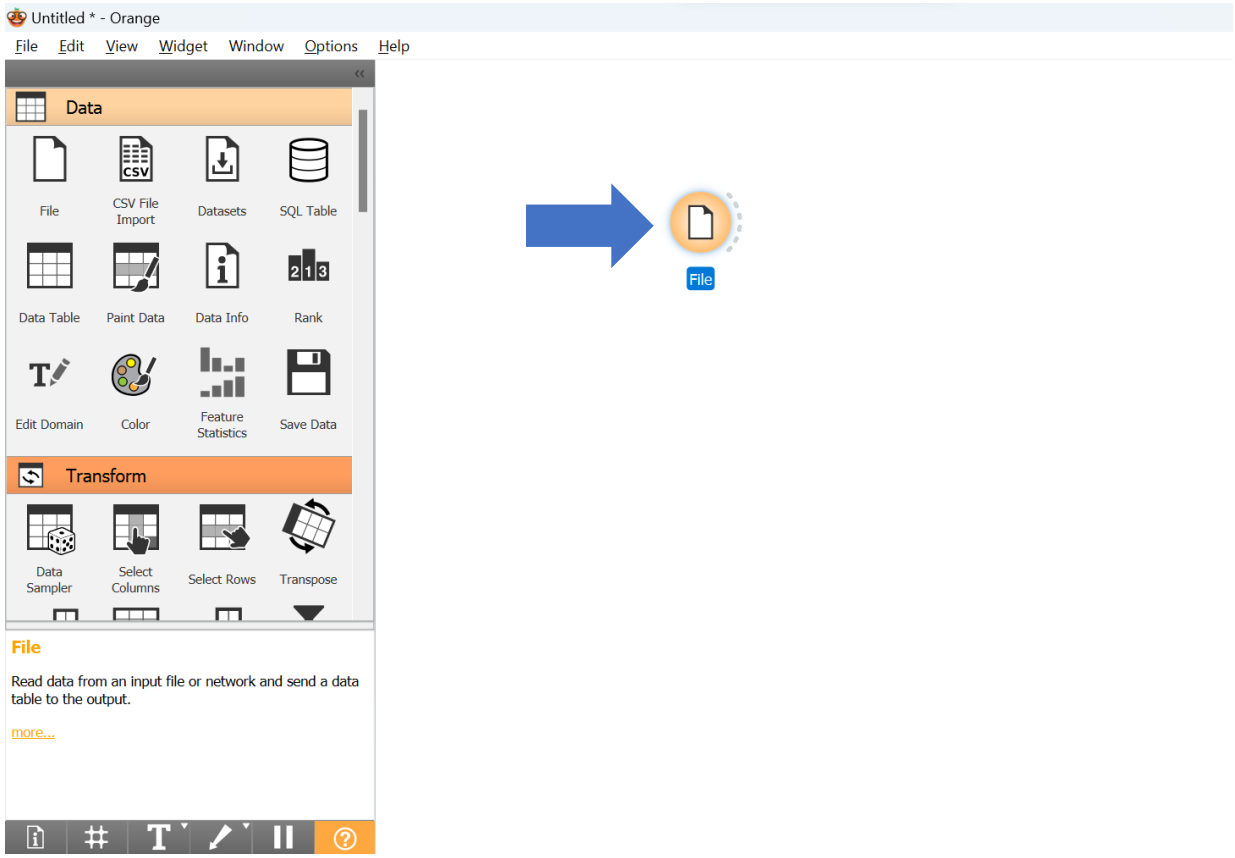
Create a New Project

- Open the Orange Data Mining application
- Welcome to Orange Page will appear
- Click on New to open a new project

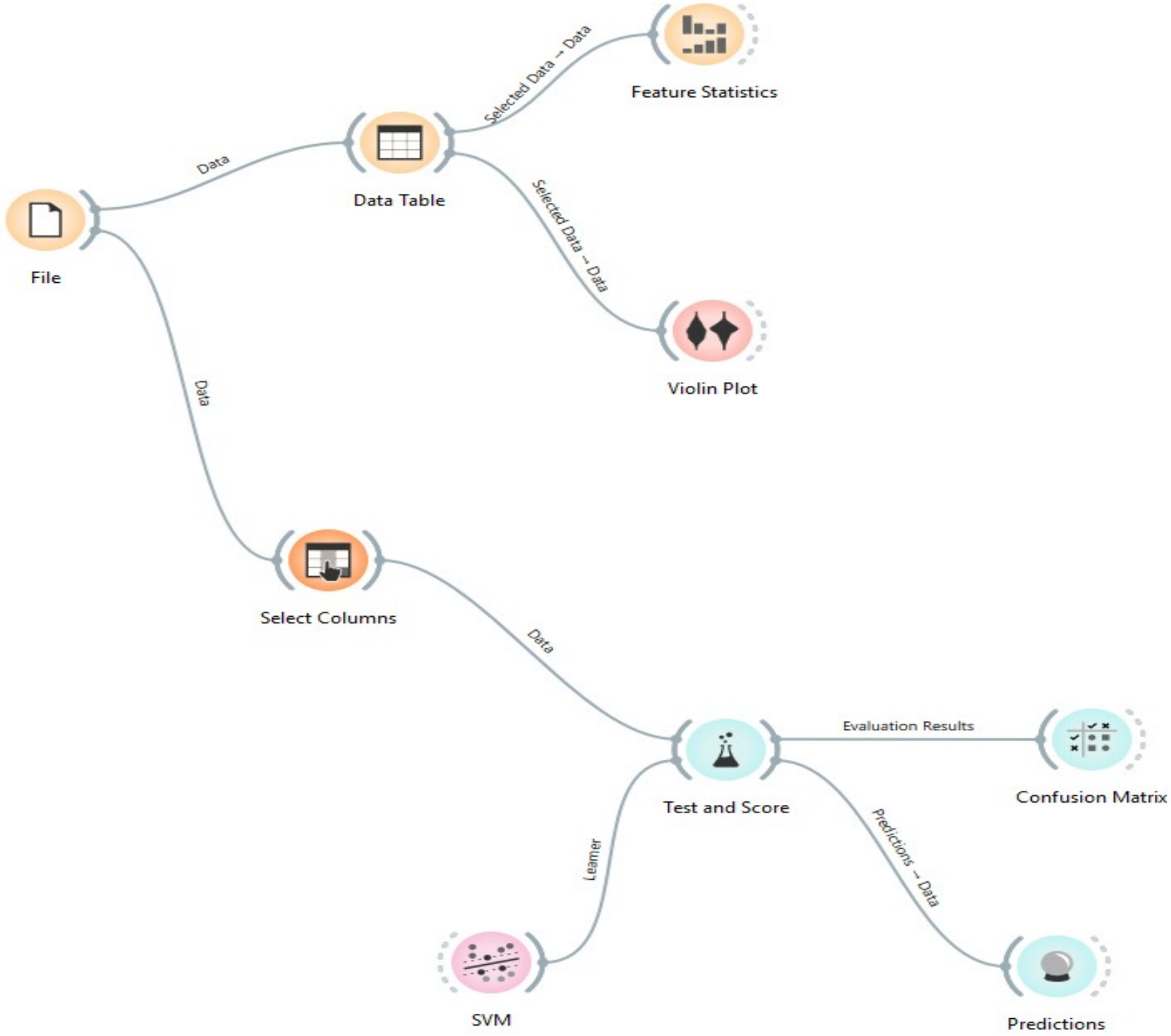


Import Dataset

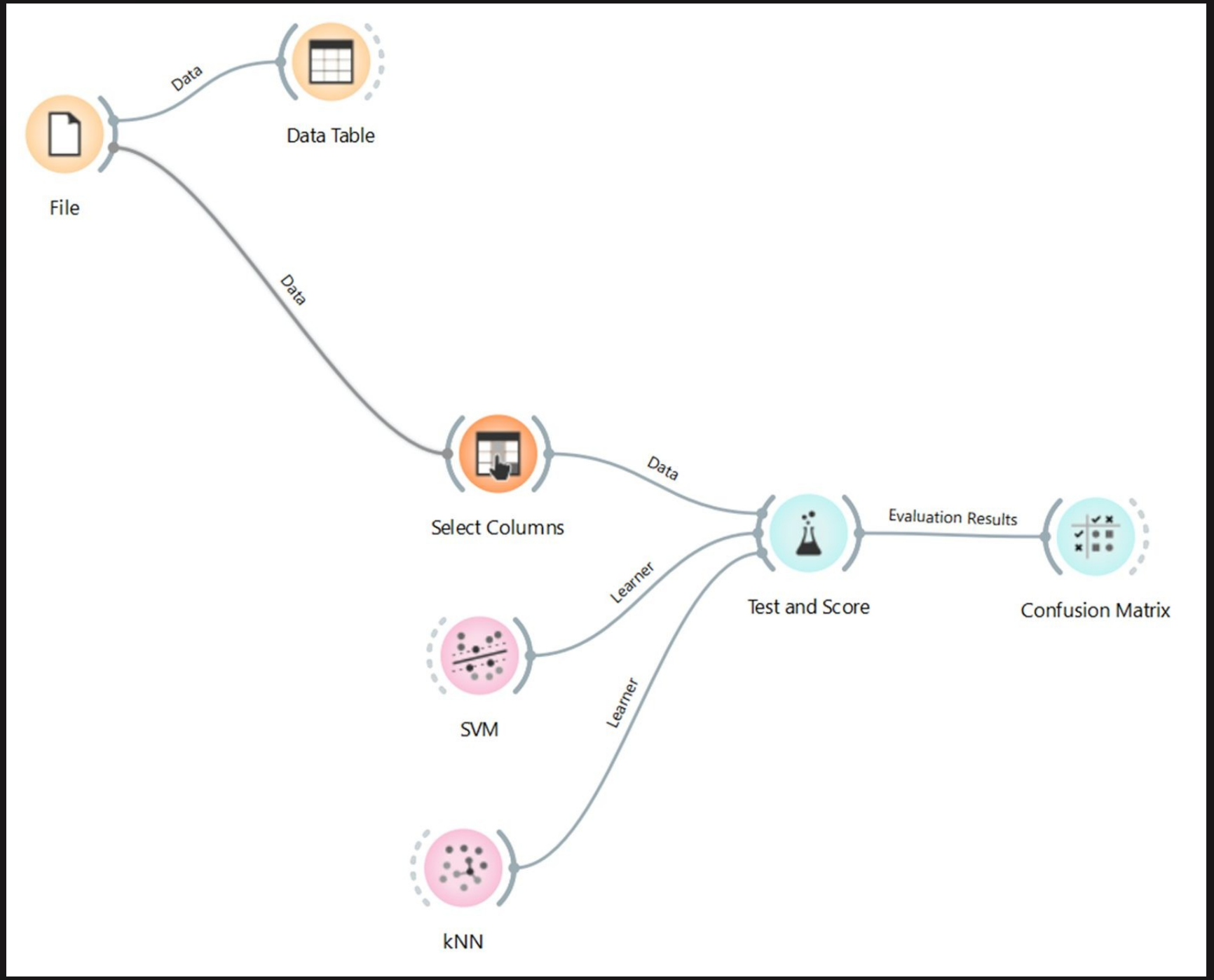
- Click on the file widget
- Grab the dataset:
- github.com/grcc-resendiz/ai-manufacturing-muskegon



Making Predictior



Orange Canvas



Demo: The Same Model in Python

- Open the Predictive Maintenance ML Demo on the course GitHub page.
- Click Open in Codespaces, open the notebook, and click Run All.
- Same idea as Orange: sensor data in, failure prediction out.
- You do not need to code. This is what a data team would build next.

Implementation and Next Steps

Technology Adoption Gap

- What technology do you already have that is underused?
  - PLC logs
  - MES systems
  - ERP data
  - Sensors
  - Maintenance logs

- Adoption is the bottleneck.

AI Readiness Reflection

- If you tried this tomorrow:

- Do you have reliable data?
- Is it centralized?
- Is it clean?
- Who owns it?

Your First AI Pilot

- Pick one problem from your Day 1 list.
- Match it to a tool: prompt, project, app, vision model, or predictive model.
- Name who owns it and what data you already have.
- Define success in numbers: minutes, scrap, or dollars.
- Commit to one step in the next 30 days.

Key Takeaways

- Workforce shortage is real.
- Technology exists.
- AI learns from patterns.
- Data quality matters.
- Adoption is leadership-driven.

Questions?


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